

Swar Mahesh Khatav

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TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript (ES6+), Go, C++, Java, SQL, Bash, HTML5, CSS3

Frontend: React.js, Next.js, TypeScript, TailwindCSS, Bootstrap, Redux, Webpack, Vite

Backend & Frameworks: Go, Node.js, Express, Django, Flask, FastAPI, REST APIs, GraphQL

Cloud & Infrastructure: Kubernetes, Docker, Google Cloud Platform (GCP), Terraform, Helm Charts

CI/CD & DevOps: Jenkins, GitHub Actions, GitLab CI, Docker, Git, Automated Pipelines, Infrastructure as Code

Databases: PostgreSQL, MongoDB, Redis, Firebase Firestore, SQL Query Optimization, Data Modeling

Testing & Quality: Jest, Pytest, JUnit, React Testing Library, Cypress, Unit Testing, Integration Testing, TDD

Methodologies: Agile/Scrum, Object-Oriented Programming, SOLID Principles, Microservices, MVC, Code Reviews

EXPERIENCE

Technical Operations Specialist | School of Social Work ASU

July 2024–Feb 2025

- Automated operational workflows and data processing pipelines using Python and Java, reducing manual processing overhead by 40% and improving data accuracy across recurring reporting and ETL tasks.
- Optimized complex SQL queries, indexing strategies, and database-driven workflows in PostgreSQL.
- Developed backend scripts and microservices that integrated external services via REST APIs, supporting scalable internal data flows, system interoperability, and reliable data exchange between operational platforms.
- Built and maintained automated data validation and transformation scripts in Python, processing thousands of records daily with structured error handling, logging, and retry logic to ensure pipeline reliability.
- Assisted with troubleshooting, debugging, and maintaining data processing workflows across multiple operational systems, diagnosing root causes in API integrations, database queries, and scheduled batch jobs.
- Created monitoring dashboards and automated alerting for pipeline health, processing throughput, and error rates, enabling the team to proactively identify and resolve issues before they impacted downstream consumers.
- Containerized Python-based automation services using Docker, standardizing development and deployment environments and enabling consistent execution across local, staging, and production setups.

Software Engineer Intern | Velocified LMS

August 2025 – May 2026

- Developed and integrated full-stack features across a scalable learning management platform using React, Go, PostgreSQL, and REST APIs, delivering end-to-end functionality from frontend UI to backend services and database layer.
- Designed and implemented backend API endpoints in Go for user authentication, session management, OTP verification, and password reset workflows, using signed cookies and database-driven architecture to ensure secure, scalable access control.
- Built role-based frontend dashboards in React with dynamic data rendering, real-time reporting, and conditional UI logic for learners, managers, coaches, and administrators, improving platform usability and operational visibility.
- Worked in Docker-based development environments and contributed to CI/CD pipelines using Git and Docker, supporting automated builds, integration testing, and reliable delivery of full-stack application updates to production.
- Debugged and resolved complex system integration issues spanning REST APIs, authentication flows, reporting modules, and user management services, collaborating with senior engineers to trace root causes across the full application stack.
- Integrated backend services with frontend components to improve data flow, streamline reporting workflows, and support real-time analytics for platform administrators and coaches.
- Wrote unit and integration tests and collaborated with QA workflows to validate functionality, catch regressions, and improve software quality and reliability before each release cycle.
- Contributed to database schema design and query optimization in PostgreSQL, improving data retrieval performance for high-traffic reporting and user management endpoints.
- Participated in Agile sprint planning, daily stand-ups, technical design discussions, and code review cycles, contributing to architecture decisions and cross-functional collaboration within a team-based engineering environment.
- Authored technical documentation including API endpoint specifications, authentication flow diagrams, and developer setup guides, supporting onboarding and knowledge transfer across the engineering team.

Project

StudySync - AI Study Platform | React, TypeScript, Python, FastAPI, Docker, GCP, Kubernetes

- Built a full-stack AI-powered study platform using React/TypeScript frontend and Python/FastAPI backend, deployed on GCP with Docker and Kubernetes for auto-scaling and high availability.
- Implemented auth-aware backend microservices for AI study plan generation, quiz creation, and real-time collaboration, processing 1,000+ daily API requests with structured error handling and retry logic.
- Designed a CI/CD pipeline using Jenkins and Docker for automated testing, containerized builds, and rolling deployments to Kubernetes, enabling rapid iteration with zero-downtime releases.

Education

Bachelor of Science in Computer Science | Arizona State University, Tempe, AZ, USA (GPA: 4.0/4.0) Aug 2022 – May 2026

Relevant Coursework: Data Structures & Algorithms, Operating Systems, Computer Networking, Distributed Systems, Database Management, Software Engineering, Cloud Computing